

EDUCATION

- **University of Technology - Vietnam National University Ho Chi Minh City** 09/2021 - 07/2025
B. E. in Engineering Physics (minor: Biomedical Engineering) (English-taught program) Ho Chi Minh City, Vietnam
 - Achieved top grades in specialized courses: Fundamentals of Quantum Mechanics and Solid State, Fundamentals of Nuclear Physics and Applications, Engineering Optics and Applications, Microprocessor Applications in Biomedical Engineering, Informatics for Engineering Physics...
 - Teaching assistant: Informatics for Engineering Physics, Laboratory of General Physics, Introduction to Artificial Intelligence
 - Thesis: Investigation and Simulation of The Mechanical Behavior of Human Skin Layers with The Hyperelastic Model using The Finite Element Method (**Grade: 9.2/10.0**)

WORK EXPERIENCES

- **AI Healthcare Co.,Ltd.** 01/2026 - Present
Vice Director - Researcher Ho Chi Minh City, Vietnam
 - Partner with international universities and companies (Japan, Taiwan, Korea,...) to become the exclusive distributor of medical AI technology in Vietnam
 - Co-organize scientific conferences and seminars on diagnostic imaging and radiotherapy (ISSRT & VART)
- **Mitalab Co.,Ltd** 06/2024 - 08/2025
Consultant Employee Ho Chi Minh City, Vietnam
 - Designed layout plans for medical equipment placement in hospitals and clinics
 - Optimized diagnostic workflows by over 50%, streamlining both staff operational procedures and equipment utilization.

RESEARCH EXPERIENCES

- **Lab of Numerical Simulation VIBE Project, HCMUT & Ireland, VNU-HCM** 10/2023 - Present
Student Researcher Ho Chi Minh City, Vietnam
 - Researched: Development of numerical methods for acoustic wave propagation problems and mathematical models of hyperelasticity for biomedical applications (**Participated in 8 research papers at 7 international conferences**)
 - Mentored 3 undergraduate students through their capstone projects
- **Department of Biomedical Engineering, HCMUT, VNU-HCM** 01/2023 - 06/2025
Student Researcher Ho Chi Minh City, Vietnam
 - Researched machine learning-based analysis and investigation of EEG signals for brain-computer interfaces (**Participated in 2 research papers and 1 coding challenge at 2 international conferences**)
 - Analyzed research data and drafted a manuscript for a peer-reviewed publication

PROJECTS

- **Secure Brain-Computer Interfaces with Homomorphic Encryption: A Case Study on Electroencephalogram Based P300 Classification** 06/2025 - 11/2025
25th Eureka Student Scientific Research Award 2025 (Semi-final Round)
 - On the bigP3BCI dataset with P300 signals, we test time-window and xDAWN features using LDA
 - Paillier HE achieves identical accuracy and AUC as plaintext, ensuring both security with an average accuracy of 85.27% and an average AUC of 57.21%
- **MEMORY: A mobile application to support memory improvement for Alzheimer's Patients** 04/2025 - 08/2025
Bach Khoa Innovation 2025 - University of Technology - VNU-HCM (Semi-final Round)
 - The survey was conducted based on the needs of nearly 100 patients, over 200 students, and more than 50 faculty members. The application includes international tests and brain games that were tested at the University.

- **TSHT: Terahertz Sensing and Healing Technology** 02/2024 - 05/2024
IU Startup Demo Day 2024 - International University - VNU-HCM - Grand-final Round
 - Rated as one of the most promising projects in the competition's final round (**Top 6 of 22**). Proposed and developed a Terahertz beam through a crystal
- **Identification and Classification of Multi-label Brain Tumors using MATLAB Software integrated with Graphical User Interface based on Convolutional Neural Network and Deep Learning Algorithms** 11/2023 - 01/2024
BME Innovation 2024 - International University - VNU-HCM - Semi-final Round
 - The brain tumor region was delineated and identified on MRI images, with an F1 result of **98.1%**, and then integrated into a GUI using MATLAB

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT

- [C.1] Nguyen, T. H., Ly, G. H., & **Hoang, D. K. D.** (2026). Leave-One-Subject-Out (LOSO)-Validated Ensemble of Bi-LSTM and Transformer Models for Unusual Activity Detection. *Journal of Physics: Conference Series*, 3180(1), 012008. <https://doi.org/10.1088/1742-6596/3180/1/012008> (SJR: 0.187, ISSN: 1742-6596).
- [C.2] **Hoang, D. K. D.**, & Thi, M. H. N. (2026). A meshfree method based on moving Kriging interpolation for the analysis of acoustic wave propagation problems. *Journal of Physics: Conference Series*, 3180(1), 012060. <https://doi.org/10.1088/1742-6596/3180/1/012060> (SJR: 0.187, ISSN: 1742-6596).
- [C.3] **Hoang, D. K. D.**, Tran, T. N., Truong, T. T., & Nguyen, T. N. (2025). Simulation of Mechanical Behavior for Human Skin Using Finite Element Method with the Third-Order Ogden Hyperelastic Model. *10th International Conference on the Development of Biomedical Engineering in Vietnam (BME 2024). IFMBE Proceedings*, vol 122. Springer, Cham. 928–939. https://doi.org/10.1007/978-3-031-90194-2_67 (SJR: 0.124, ISBN: 978-3-031-90193-5)
- [C.4] Manh, H. T., & **Duy, K. D. H.** (2023). Harnessing the Self-Healing Effect of Terahertz Rays: Advancements and Applications. *2023 1st International Conference on Health Science and Technology (ICHST). IEEE Xplore*. 1-6. <https://doi.org/10.1109/ichst59286.2023.10565312> (ISBN: 979-8-3503-1572-1)

HONORS & AWARDS

- “Student with 5 good criteria 2025” title - National level 01/2026
Central Committee of Viet Nam National Union of Students
- “Advanced youth follow President Ho Chi Minh’s wills 2025” title 05/2025
University of Technology - Vietnam National University Ho Chi Minh City
- OISP Talented Students 2023-2024 03/2025
Office for International Study Programs - University of Technology - Vietnam National University Ho Chi Minh City
- The Potential Project (Grand-final Round) at IU Startup Demo Day 2024 05/2024
International University - Vietnam National University Ho Chi Minh City
- Second Prize at the 2022 Engineering Fair, Round 2 01/2022
Faculty of Applied Science - University of Technology - Vietnam National University Ho Chi Minh City

SKILLS

- **Languages:** MATLAB, Python
- **Tools:** Numpy, Tensorflow, Keras, Pytorch, Scikit-Learn, Matplotlib, Pandas, LaTeX
- **Specialization:** Machine Learning, Deep Learning, AI and Data Science in Health, Numerical Analysis
- **Spoken languages:** Vietnamese (Native), English (Proficient), Japanese (Basic)

CERTIFICATIONS

- Certification of participated in the Winter School on Quantum Computing from November 21-23, 2025 at the University of Technology - Vietnam National University Ho Chi Minh City 23/11/2025
Vietnam Institute for Advanced Study in Mathematics (VIASM)
- Certification of Top 40 Bach Khoa Innovation 2025 19/09/2025
Ho Chi Minh City University of Technology - Vietnam National University
- Certification of Won third prize in the TOFAS 2024 programming competition - Amateur group 23/11/2024
Ho Chi Minh City University of Technology - Vietnam National University
- Certification of completion the Online Continuing Professional Development Program: The 33rd Online Continuing Professional Development Platform for Asian Radiologic Technologist's Societies and The 34th Continuing Professional Development (CPD) Program of Macao Radiological Technologists' Association 24/01/2024
Macao Radiological Technologists' Association
- Certification of successfully completed the courses of 2022 Holiday School-Introduction to Biomedical Mechanics from June 20, 2022 to June 27, 2022 29/06/2022
Department of Biomedical Engineering, National Cheng Kung University